## Economics Basics: Introduction

*Economics may appear to be the study of complicated tables and charts, statistics and numbers, but, more specifically, it is the study of what constitutes rational human behaviour in the endeavour to fulfil needs and wants.*

*economics makes the assumption that human beings will aim to fulfill their self-interests. It also assumes that individuals are rational in their efforts to fulfill their unlimited wants and needs. Economics, therefore, is a social science, which examines people behaving according to their self-interests. The definition set out at the turn of the twentieth century by Alfred Marshall, author of "The Principles of Economics", reflects the complexity underlying economics: "Thus it is on one side the study of wealth; and on the other, and more important side, a part of the study of man."*

**Scarcity**

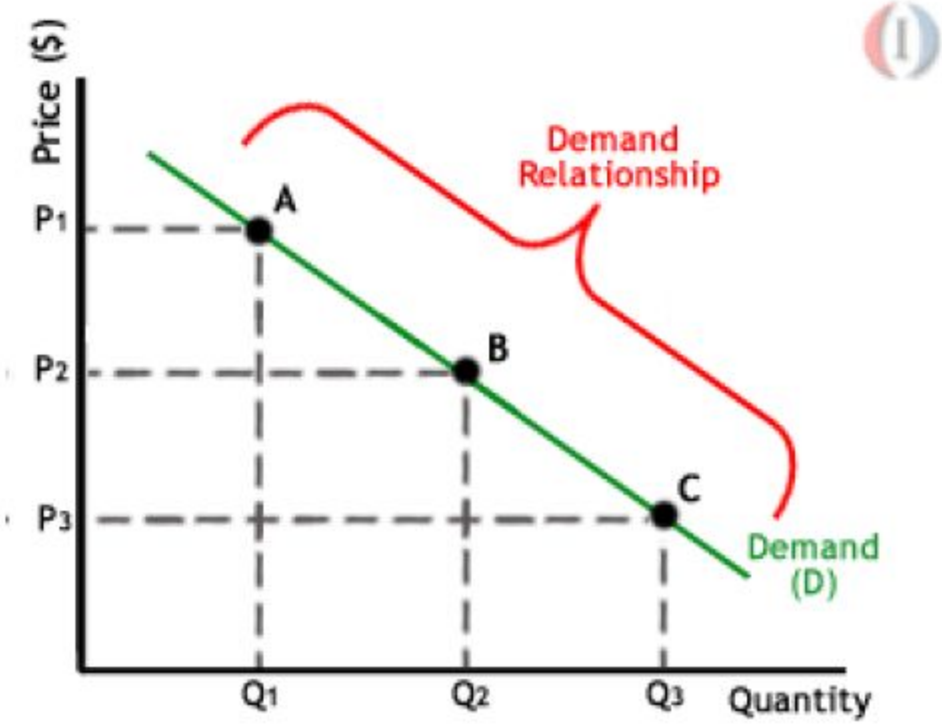
*Scarcity, a concept we already implicitly discussed in the introduction to this tutorial, refers to the tension between our limited resources and our unlimited wants and needs. For an individual, resources include time, money and skill. For a country, limited resources include natural resources, capital, labour force and technology. Because all of our resources are limited in comparison to all of our wants and needs, individuals and nations have to make decisions regarding what goods and services they can buy and which ones they must forgo. For example, if you choose to buy one DVD as opposed to two video tapes, you must give up owning a second movie of inferior technology in exchange for the higher quality of the one DVD. Of course, each individual and nation will have different values, but by having different levels of (scarce) resources, people and nations each form some of these values as a result of the particular scarcities with which they are faced. So, because of scarcity, people and economies must make decisions over how to allocate their resources. Economics, in turn, aims to study why we make these decisions and how we allocate our resources most efficiently*

***Supply and demand*** *is perhaps one of the most fundamental concepts of economics and it is the backbone of a market economy. Demand refers to how much (quantity) of a product or service is desired by buyers. The quantity demanded is the amount of a product people are willing to buy at a certain price; the relationship between price and quantity demanded is known as the demand relationship. Supply represents how much the market can offer. The quantity supplied refers to the amount of a certain good producers are willing to supply when receiving a certain price. The correlation between price and how much of a good or service is supplied to the market is known as the supply relationship. Price, therefore, is a reflection of supply and demand.*

*The relationship between demand and supply underlie the forces behind the allocation of resources. In market economy theories, demand and supply theory will allocate resources in the most efficient way possible. How? Let us take a closer look at the law of demand and the law of supply*

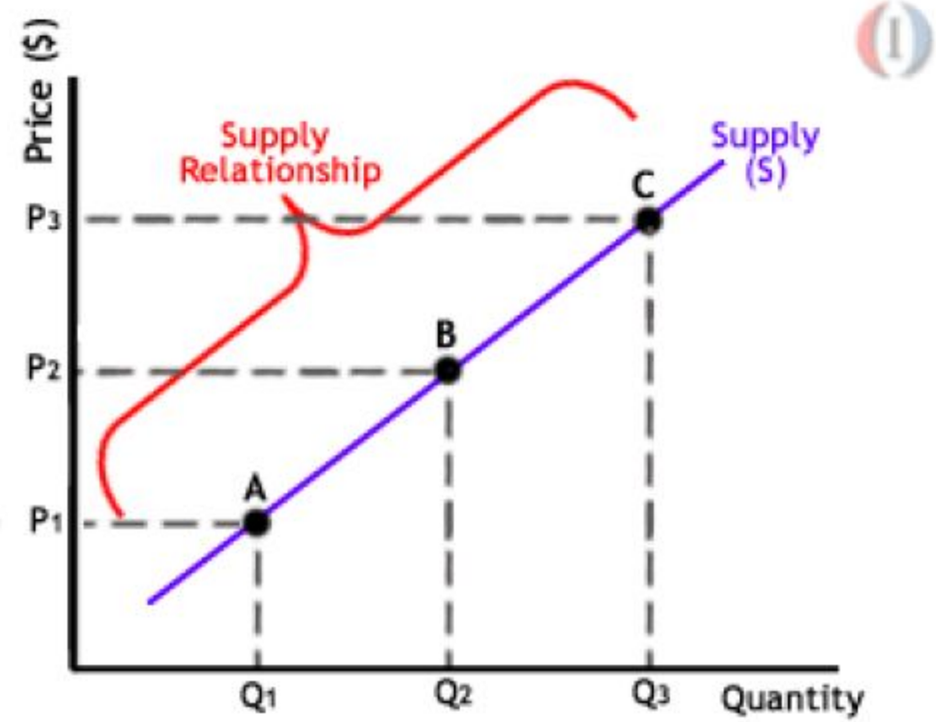
# The Law of Demand

*The law of demand states that, if all other factors remain equal, the higher the price of a good, the less people will demand that good. In other words, the higher the price, the lower the quantity demanded. The amount of a good that buyers purchase at a higher price is less because as the price of a good goes up, so does the opportunity cost of buying that good. As a result, people will naturally avoid buying a product that will force them to forgo the consumption of something else they value more. The chart below shows that the curve is a downward slope. The higher the price of a good the lower the quantity demanded (A), and the lower the price, the more the good will be in demand (C).*

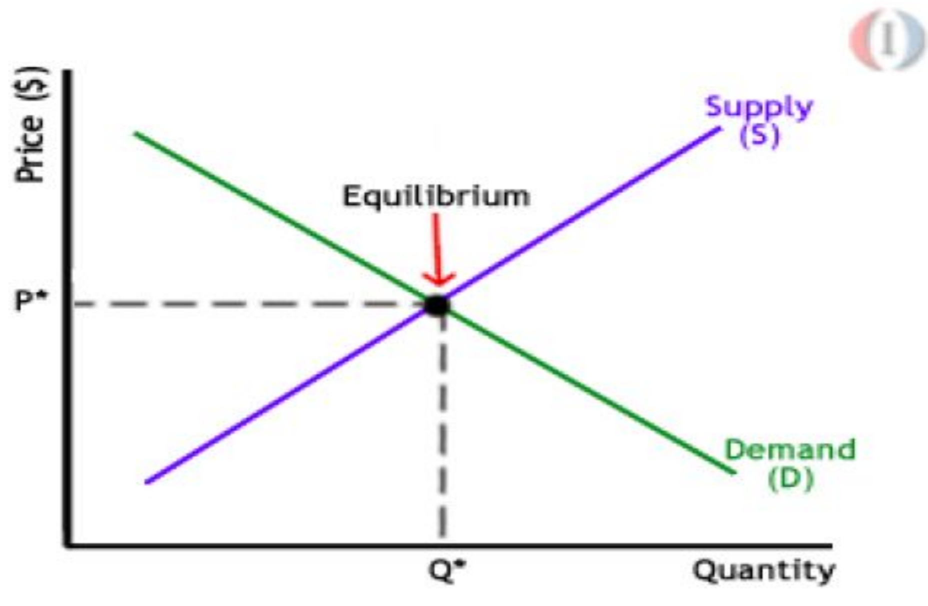


***The Law of Supply***

*Like the law of demand, the law of supply demonstrates the quantities that will be sold at a certain price. But unlike the law of demand, the supply relationship shows an upward slope. This means that the higher the price, the higher the quantity supplied. Producers supply more at a higher price because selling a higher quantity at a higher price increases revenue.*



*When supply and demand are equal (i.e. when the supply function and demand function intersect) the economy is said to be at equilibrium. At this point, the allocation of goods is at its most efficient because the amount of goods being supplied is exactly the same as the amount of goods being demanded. Thus, everyone (individuals, firms, or countries) is satisfied with the current economic condition. At the given price, suppliers are selling all the goods that they have produced and consumers are getting all the goods that they are demanding. As you can see on the chart, equilibrium occurs at the intersection of the demand and supply curve, which indicates no allocative inefficiency. At this point, the price of the goods will be P* and the quantity will be Q*. These figures are referred to as equilibrium price and quantity.*



***What is the 'Time Value of Money - TVM'***

*The time value of money (TVM) is the concept that money available at the present time is worth more than the identical sum in the future due to its potential earning capacity. This core principle of finance holds that, provided money can earn interest, any amount of money is worth more the sooner it is received. TVM is also sometimes referred to as present discounted value.*

*The time value of money is an important concept not just for individuals, but also for making business decisions. Companies consider the time value of money in making decisions about investing in new product development, acquiring new business equipment or facilities, and in establishing credit terms for the sale of their products or services.*

*A specific formula can be used for calculating the future value of money so that it can be compared to the present value:*

$$FV = PV \times [1 + (i / n)]^{(n \times t)}$$

**FV** = the future value of money

**PV** = the present value

**i** = the interest rate or other return that can be earned on the money

**t** = the number of years to take into consideration

**n** = the number of compounding periods of interest per year

**What is Demand?**

*Demand for a commodity refers to the quantity of the commodity that people are willing to purchase at a specific price per unit of time, other factors (such as price of related goods, income, tastes and preferences, advertising, etc) being constant. Demand includes the desire to buy the commodity accompanied by the willingness to buy it and sufficient purchasing power to purchase it. For instance-Everyone might have willingness to buy "Mercedes-S class" but only a few have the ability to pay for it. Thus, everyone cannot be said to have a demand for the car "Mercedes-s Class".*

*Demand curve is a diagrammatic representation of demand schedule. It is a graphical representation of price- quantity relationship. Individual demand curve shows the highest price which an individual is willing to pay for different quantities of the commodity. While, each point on the market demand curve depicts the maximum quantity of the commodity which all consumers taken together would be willing to buy at each level of price, under given demand conditions.*

*Demand curve has a negative slope, i.e, it slopes downwards from left to right depicting that with increase in price, quantity demanded falls and vice versa.*

***The reasons for a downward sloping demand curve can be explained as follows-***

***Income effect-*** *With the fall in price of a commodity, the purchasing power of consumer increases. Thus, he can buy same quantity of commodity with less money or he can purchase greater quantities of same commodity with same money. Similarly, if the price of a commodity rises, it is equivalent to decrease in income of the consumer as now he has to spend more for buying the same quantity as before. This change in purchasing power due to price change is known as income effect.*

***Substitution effect-*** *When price of a commodity falls, it becomes relatively cheaper compared to other commodities whose price have not changed. Thus, the consumer tend to consume more of the commodity whose price has fallen, i.e, they tend to substitute that commodity for other commodities which have not become relatively dear.*

***Law of diminishing marginal utility-*** *It is the basic cause of the law of demand. The law of diminishing marginal utility states that as an individual consumes more and more units of a commodity, the utility derived from it goes on decreasing. So as to get maximum satisfaction, an individual purchases in such a manner that the marginal utility of the commodity is equal to the price of the commodity. When the price of commodity falls, a rational consumer purchases more so as to equate the marginal utility and the price level. Thus, if a consumer wants to purchase larger quantities, then the price must be lowered. This is what the law of demand also states.*

***Exceptions to Law of Demand***

*The instances where law of demand is not applicable are as follows-*

*There are certain goods which are purchased mainly for their snob appeal, such as, diamonds, air conditioners, luxury cars, antique paintings, etc. These goods are used as status symbols to display one's wealth. The more expensive these goods become, more valuable will be they as status symbols and more will be there demand. Thus, such goods are purchased more at higher price and are purchased less at lower prices. Such goods are called as **conspicuous goods**.*

*The law of demand is also not applicable in case of **giffen goods**. Giffen goods are those inferior goods, whose income effect is stronger than substitution effect. These are consumed by poor households as a necessity. For instance, potatoes, animal fat oil, low quality rice, etc. An increase in price of such good increases its demand and a decrease in price of such good decreases its demand.*

*The law of demand does not apply in case of expectations of change in price of the commodity, i.e, in case of **speculation**. Consumers tend to purchase less or tend to postpone the purchase if they expect a fall in price of commodity in future. Similarly, they tend to purchase more at high price expecting the prices to increase in future*.

# Utility

*Underlying the laws of demand and supply is the concept of utility, which represents the advantage or fulfilment a person receives from consuming a good or service. Utility, then, explains how individuals and economies aim to gain optimal satisfaction in dealing with scarcity.*

*Utility is an abstract concept rather than a concrete, observable quantity. The units to which we assign an "amount" of utility, therefore, are arbitrary, representing a relative value. Total utility is the aggregate sum of satisfaction or benefit that an individual gains from consuming a given amount of goods or services in an economy. The amount of a person's total utility corresponds to the person's level of consumption. Usually, the more the person consumes, the larger his or her total utility will be. Marginal utility is the additional satisfaction, or amount of utility, gained from each extra unit of consumption.*

*Although total utility usually increases as more of a good is consumed, marginal utility usually decreases with each additional increase in the consumption of a good. This decrease demonstrates the law of diminishing marginal utility. Because there is a certain threshold of satisfaction, the consumer will no longer receive the same pleasure from consumption once that threshold is crossed. In other words, total utility will increase at a slower pace as an individual increases the quantity consumed.*

*Take, for example, a chocolate bar. Let's say that after eating one chocolate bar your sweet tooth has been satisfied. Your marginal utility (and total utility) after eating one chocolate bar will be quite high. But if you eat more chocolate bars, the pleasure of each additional chocolate bar will be less than the pleasure you received from eating the one before - probably because you are starting to feel full or you have had too many sweets for one day.*

*This table shows that total utility will increase at a much slower rate as marginal utility diminishes with each additional bar. Notice how the first chocolate bar gives a total utility of 70 but the next three chocolate bars together increase total utility by only 18 additional units.*

| Chocolate Bars Eaten | Marginal Chocolate Utility | Total Chocolate Utility |
|---|---|---|
| 0 | 0 | 0 |
| 1 | 70 | 70 |
| 2 | 10 | 80 |
| 3 | 5 | 85 |
| 4 | 3 | 88 |

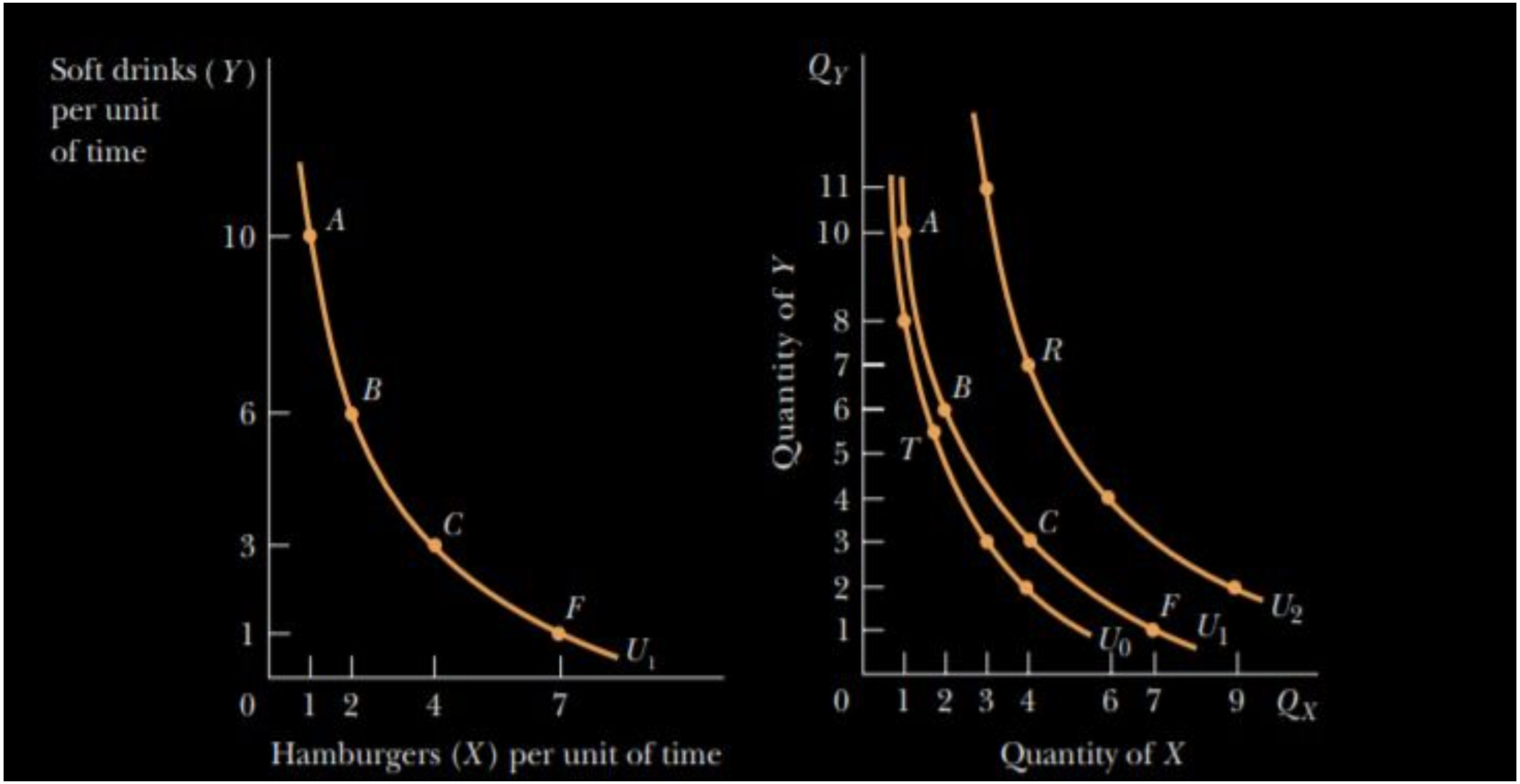
***The law of diminishing marginal utility helps economists understand the law of demand and the negative sloping demand curve. The less of something you have, the more satisfaction you gain from each additional unit you consume; the marginal utility you gain from that product is therefore higher, giving you a higher willingness to pay more for it. Prices are lower at a higher quantity demanded because your additional satisfaction diminishes as you demand more.***

# *Consumer Preferences and Choices*

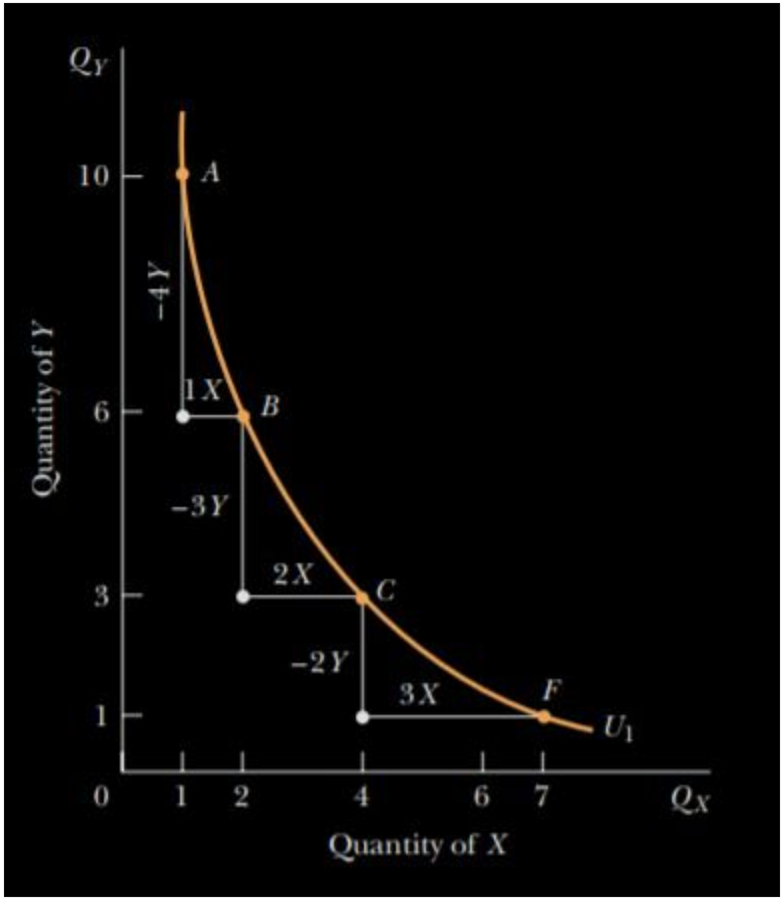
## *Utility Analysis*

*The level of happiness or satisfaction that a person receives from his or her circumstances. Utility is a measure of well-being and, according to utilitarian's, is the ultimate objective of all public and private actions. The proper goal of the government, they claim, is to maximize the sum of utility achieved by everyone.*

# Consumer Taste - Indifference Curves



# The Marginal Rate of Substitution

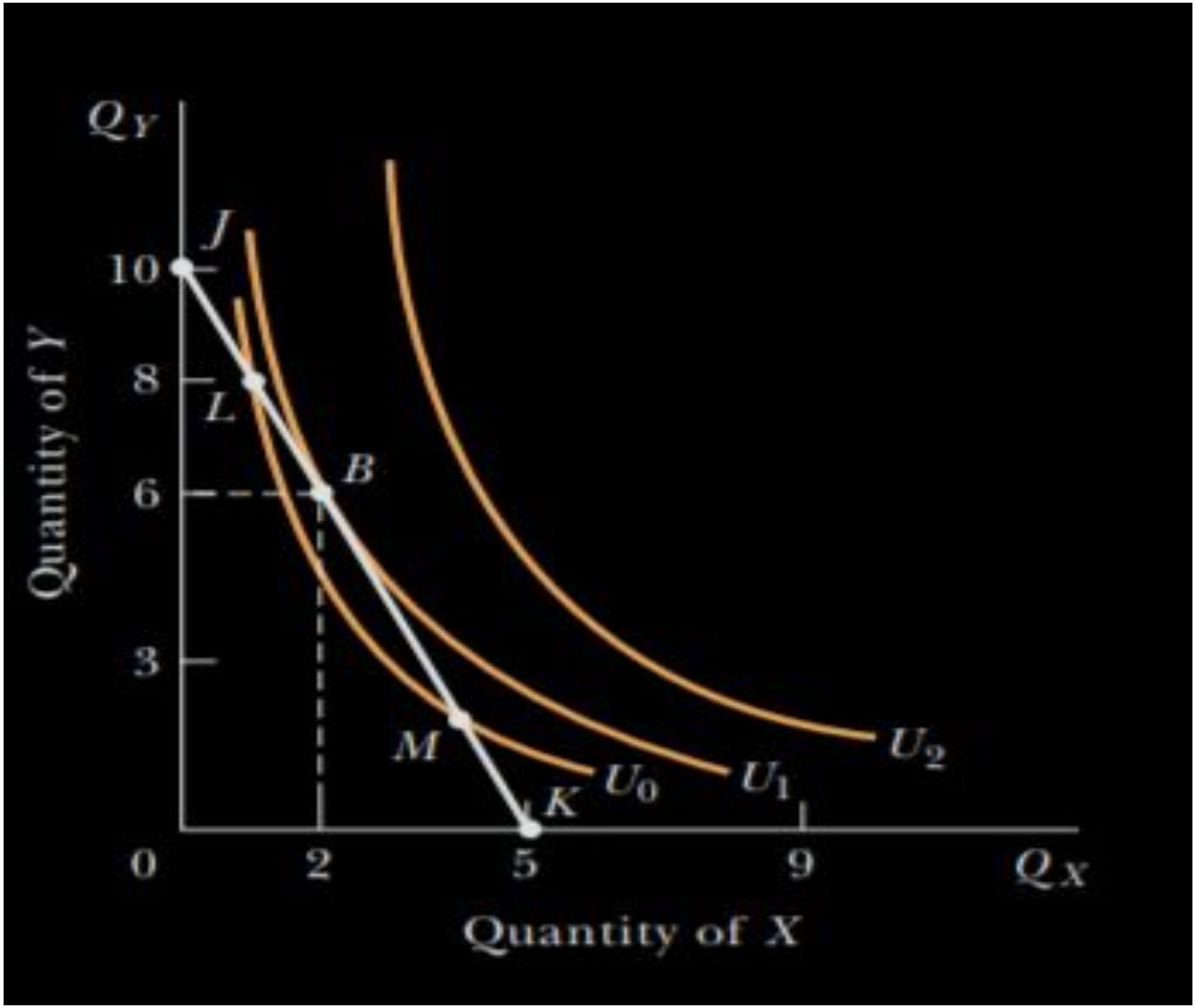



***Refers to the amount of one good that an individual is willing to give up for an additional unit of another good while maintaining the same level of satisfaction or remaining on the same indifference curve.***

# Consumer's Choice - Utility Maximization

*The rational consumer seeks to maximize the utility or satisfaction received in spending his or her income. A rational consumer maximizes utility by trying to attain the highest indifference curve possible, given his or her budget line. This occurs where an indifference curve is tangent to the budget line so that the slope of the indifference curve is equal to the slope of the budget line.*

$$\mathbf{MRS_{XY} = PX/PY}$$



$Q_Y$
Quantity of $Y$
10
8
6
3
0
J
L
B
M
K
$U_0$
$U_1$
$U_2$
2
5
9
$Q_X$
Quantity of $X$